Modern Physics 2
Nuclear Physics

Nucleus:
→ It is center of atom
→ It's radius is in order of fermi ($1\ Fm = 10^{-15} m$), while the radius of the atom is of the order of angstroms
→ It has all the +ve charge of atom.
→ It has practically all the mass of atom.
→ It was discovered by Rutherford → By α-scattering experiment (1911).
→ Nucleus contains of Neutrons and protons.

Neutrons:
→ Discovered by James Chadwick in 1932
→ Electrically neutral
→ mass = 1.6749×10^{-27} kg = 1 amu
→ Representation: ${}_0n^1$ → mass {in amu}
↳ charge {in e}

Protons:
→ Discovered by Goldstein
→ Positively charged
→ Charge = $Q = +e = 1.6 \times 10^{-19} C$
→ mass = 1.6726×10^{-27} kg ≈ 1 amu
→ Representation: ${}_1p^1$

electron: ${}_{-1}e^0$

mass of $e^\ominus$ in amu = $\frac{1\ amu}{1800} \approx 0$

Positron: ${}_{+1}e^0$

$Q = (+1)\ 1.6 \times 10^{-19} C$

$m = 0$ amu $= \frac{9.1 \times 10^{-31}}{1.66 \times 10^{-27}}$ amu

Nuclei: Plural of Nucleus
Protons + Neutrons = Nucleons

Representation of Nucleus:

$$\boxed{{}_Z X^A}$$

X → Symbol of Nucleus
Z = No. of protons = atomic number
If atom is neutral ⇒ Z = no. of $e^\ominus$
No. of neutrons = (A − Z)
A = no. of nucleons = no. of protons + no. of neutrons = mass no.

Ex. ${}_7N^{14}$

N → Nitrogen
No. of protons = 7
No. of nucleons = 14
No. of neutrons = 14 − 7 = 7
Charge on nucleus = 7e
Mass = 14 amu

${}_{18}Ar^{(40)}$: Ar = Argon
no. of proton = 18
no. of nucleons = 40 = 18 + no. of neutron
no. of neutrons = 22

Classification of Nuclei:

① **Isotopes**: If nuclei have same atomic no. (Z), but different mass no (A).

	${}_6C^{12}$	${}_6C^{13}$	${}_6C^{14}$
Z =	6	6	6
No. of nucleons =	12	13	14
No. of neutrons =	6	7	8

② **Isobars**: If nuclie have same no. (A) but different atomic (Z).

	${}_{16}S^{40}$	${}_{17}Cl^{40}$	${}_{18}Ar^{40}$	${}_{19}K^{40}$
Z =	16	17	18	19
No. of nucleons =	40	40	40	40
No. of neutrons =	24	23	22	21

③ **Isotones**: Nuclie having same number of neutrons (A−Z) are isotones of each other.

	${}_6C^{12}$	${}_7N^{13}$	${}_8O^{14}$
Z =	6	7	8
No. of nucleons =	12	13	14
No. of neutrons =	6	6	6

④ **Isodiaphers**: Nuclie having equal difference b/w no. of neutrons and no. of protons are isodiaphers of each other.

$$(A-Z)-Z = A-2Z$$

	${}_{92}U^{238}$	${}_{90}Th^{234}$
Z =	92	90
No. of nucleons =	238	234
No. of neutrons =	146	144
No. of neutrons − Z =	54	54

Size of Nucleus:

→ We found that nucleus has same density

$V \propto m$

$\Rightarrow V \propto$ Mass No. (A)

$= \frac{4}{3}\pi R^3 \propto A$

$\boxed{R \propto A^{1/3}}$

$\star \boxed{R = R_0 A^{1/3}}$

emperical

R_0 = Empirical Constant

$= 1.2 \times 10^{-15} m = 1.2 fm$

Q Find radius of Al nucleus. $_{13}Al^{27}$

$R = R_0 (27)^{1/3} = 3R_0 = 3 \times 1.2 = 3.6 fm$

Density of Nucleus:

$\rho = \frac{m}{V} = \frac{Zm_p + (A-Z)m_n}{\frac{4}{3}\pi R^3} = \frac{Zm_p + Am_n - Zm_n}{\frac{4}{3}\pi R^3}$ $\{m_p \approx m_n\}$

$= \rho = \frac{3Am_n}{4\pi (R_0 A^{1/3})^3} = \frac{3Am_n}{4\pi R_0^3 A} = \frac{3m_n}{4\pi R_0^3} = \frac{3 \times 1.67 \times 10^{-27}}{4 \times 3.14 \times (1.2 \times 10^{-15})^3}$

$\star \boxed{\rho = 2.3 \times 10^{17} kg/m^3}$ very large

Q If earth is made up of only nuclei. what should be its radius?

$m_{earth} = 6 \times 10^{24} kg$

$\rho V = m \Rightarrow 2.3 \times 10^{17} \times \frac{4}{3}\pi R^3 = 6 \times 10^{24}$

$R^3 = \frac{6 \times 3 \times 10^7}{2.3 \times 4 \times 3.14} = 6.2 \times 10^6 \Rightarrow R = (6.2)^{1/3} \times 100 = 1.83 \times 100 = 183 m$

$\therefore \boxed{R = 183 m}$ on the other hand, the actual radius of earth is 6400km = 64,00,000m

Average mass of nuclei:

Ex

	Cl^{35}	Cl^{37}
	80%	20%
No. of moles (n) =	80	20

assuming total moles to be 100

$\boxed{\text{Avg atomic mass} = \frac{n_1 A_1 + n_2 A_2}{n_1 + n_2} = \frac{\text{mass (total)}}{\text{total moles}}}$

$\text{Avg atomic mass} = \frac{80 \times 35 + 20 \times 37}{100}$

$= \frac{2800 + 740}{100}$

Avg atomic mass = 35.4 amu

Energy mass Equivalence:

→ Energy can be converted to mass and mass can be converted in energy.

Energy ⟷ mass

$$E = mc^2$$

m = mass (in kg)
E = energy
c = speed of light in vacuum.

Q Find energy corresponding to 1 amu mass.

$$E = mc^2 = (1.66\times10^{-27}) \times (3\times10^8)^2 \text{ J}$$
$$= 1.66\times9\times10^{-11} \text{ J}$$

∴ 1 amu ⇒ 931.5 MeV

$$E = \frac{1.66\times9\times10^{-11}}{1.6\times10^{-19}} \text{ eV} = 9.315\times10^8 \text{ eV}$$

Q Find energy corresponding to mass of 1 electron

$$E_0 = mc^2 = (9.1\times10^{-31} \text{ kg})(3\times10^8 \text{ m/s})^2 = 9.1\times9\times10^{-15} \text{ J} = \frac{9.1\times9\times10^{-15}}{1.67\times10^{-19}} \text{ eV}$$

$$E = \frac{9.1\times9}{1.6} \times 10^4 \text{ eV} = 51.1\times10^4 \text{ eV} = 511\times10^3 \text{ eV} = 511 \text{ keV} = 0.511 \text{ MeV}$$

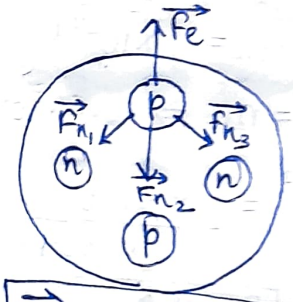
$$E = 0.511 \text{ MeV}$$

Nuclear Force:

→ Doesn't depend on charge
→ It acts b/w nucleons ie p-p, p-n, n-n
→ It is an attractive force
→ Nuclear force is very short range force

$$F \propto \frac{1}{r^{(\text{High Power})}}$$

He-Nucleus

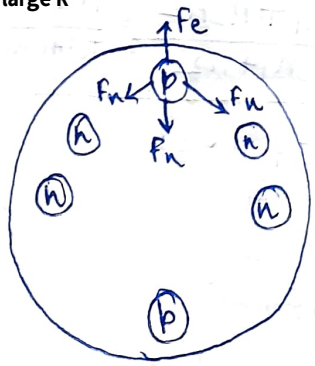


→ Nuclear force is responsible for stability of nucleus.

$$\vec{F}_{nuclear} = \vec{F}_{n_1} + \vec{F}_{n_2} + \vec{F}_{n_3}$$

→ Heavier nuclei have more no. of neutrons than proton.

implies large R



In heavy nucleus Fe has more effect on Fn as radius becomes large. So to balance Fe, no. of neutrons must be more so as to increase the attractive nuclear force.

* If heavy, large nucleus is unstable, then it will disintegrate into smaller nuclei and hence gets stable. This process is known as radioactivity.

Mass defect: It is difference of mass b/w masses of nucleons and mass of nucleus.

$$_{Z}X^{A} \longrightarrow Z\,{}_{1}p^{1} + (A-Z)\,{}_{0}n^{1}$$

nucleus

m_p = mass of proton
m_n = mass of neutron
m_x = mass of X-nucleus
M_H = mass of H-atom
m_e = mass of $e^{\ominus}$
M_x = mass of X-atom

$$\Delta m = \{Zm_p + (A-Z)m_n\} - \{m_x\}$$

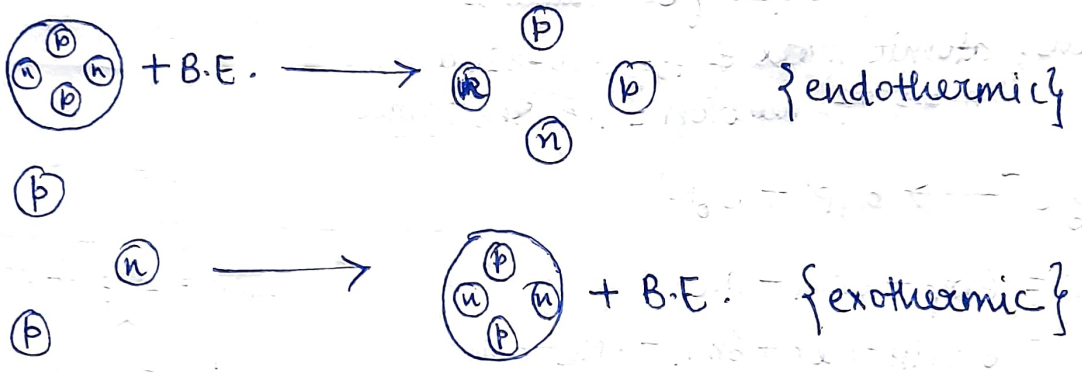
$$\Delta m = \{Z\{M_H - m_e\} + (A-Z)m_n\} - \{M_x - Zm_e\}$$

$$\Delta m = ZM_H + (A-Z)m_n - M_x$$

Binding Energy {B.E.} of nucleus:

→ B.E. is minimum energy required to break a nucleus against its attractive forces into constituent nucleons.

→ If we make a nucleus from constituent nucleons, then some energy will be liberated, this energy will be equal to B.E.



LHS ⇒ mass more

RHS ⇒ Energy more

* $B.E. = (\Delta m)c^2 = (\text{mass defect})c^2$

* $B.E. = (\Delta m \text{ in amu}) \times 931.5 \text{ MeV}$

Stability of nucleus:

$$\text{Stability} \propto \frac{\text{Binding Energy}}{\text{No. of nucleons}}$$

$$\text{Stability} \propto \frac{B.E.}{A}$$

Ex

	B.E. (MeV)	A	BE/A
X	100	10	10
Y	60	5	12
Z	80	10	8

Stability order ⇒ B > A > C

Q Find ① Mass defect
② Binding Energy
③ $\frac{B.E.}{A}$

for α-particle ie He nucleus

Given,
atomic mass of He = 4.002600 amu
atomic mass of H = 1.007826 amu
mass of neutron = 1.008665 amu

① $_2He^4 \longrightarrow 2_1p^1 + 2_0n^1$

$\Delta m = 2m_p + 2m_n - m_{He}$
$= 2\{M_H - m_e\} + 2m_n - \{M_{He} - 2m_e\}$
$= 2M_H + 2m_n - M_{He}$
$= 2\times1.007826 + 2\times1.008665 - 4.002600$
$= 0.030382$ amu

② $B.E. = \Delta mc^2$
$BE = 0.030382 \times 931.5$ MeV
$BE = 28.300$ MeV

③ $\frac{BE}{A} = \frac{28.300}{4} = 7.075$ MeV

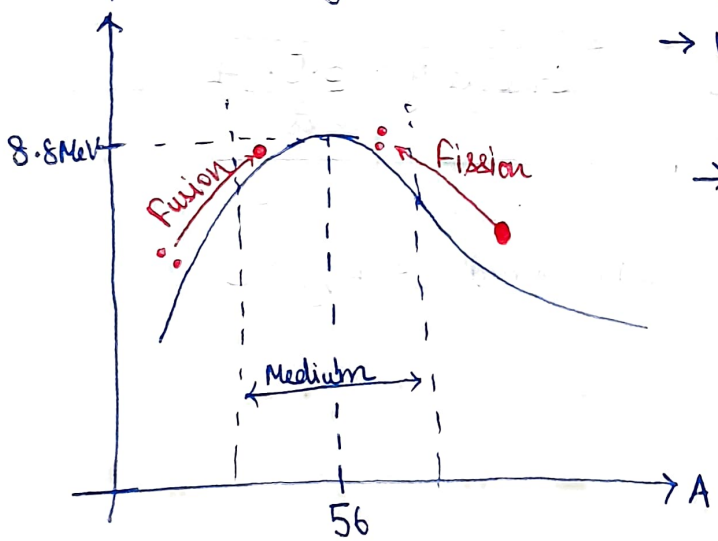
Q Find mass defect for $_6C^{12}$ nucleus.
Given, atomic mass of H = 1.007826 amu
mass of neutron = 1.008665 amu

$_6C^{12} \longrightarrow 6_1p^1 + 6_0n^1$

$\Delta m = 6m_p + 6m_n - m_C$
$= 6\{M_H - m_e\} + 6m_n - \{M_C - 6m_e\}$
$= 6M_H + 6m_n - M_C$
$= 6\times1.007826 + 6\times1.008665 - 12$
$\Delta m = 0.098946$ amu

1 amu = $\frac{1}{12^{th}}$ of atomic mass of C
$M_C = 12$ amu

Note:
BE/A ie Stability



→ $_{26}Fe^{56}$ is most stable nucleus
→ Medium sized nuclei are more stable as compared to lighter and heavier nuclei.
→ Very lighter nuclei may combine and merge to form a bigger nucleus having more BE/A and hence more stable. This process is nuclear fusion.
→ Very heavy nucleus may disintegrate and break into lighter nuclei to have more BE/A and hence more stable. This process is nuclear fission.

Fusion: $X + Y \longrightarrow Z + \text{Energy}$

BE = B_1 B_2 B_3

Energy liberate = $B_3 - (B_1 + B_2)$

Fission: $A \longrightarrow C + D + \text{Energy}$

BE = B_1 B_2 B_3

Energy liberate = $E = (B_2 + B_3) - B_1$

* In these nuclear reaction, energy liberated is always equal to change in total binding energy $\{B_{product} - B_{Reactant}\}$

Ex $_{X}^{m}R$ = $W^{*}X \longrightarrow Y + Z$

B_1 B_2 B_3 B_4 $\quad E = (B_3 + B_4) - (B_1 + B_2)$

Radioactivity:

(X) Unstable nucleus $\longrightarrow$ (Y) Stable nucleus + Radiations (α, β, γ)

→ Spontaneous conversion of unstable nucleus into stable nucleus, while emitting α, β, γ particles is known as radioactivity.

→ Any substance which shows radioactivity is known as radioactive substance.

α-particle
→ He nucleus
He^{+2} or $_2He^4$

β^- particle
→ electron
→ $_{-1}e^0 = {}_{-1}\beta^0$
→ Charge = $-e = -1.6 \times 10^{-19} C$
→ Mass = 9.1×10^{-31} kg $\approx \frac{1}{1800}$ amu

β^+ particle
→ Positron
→ $_{+1}e^0 = {}_{1}\beta^0$
→ $Q = +e$
→ Mass = 9.1×10^{-31} kg

γ particle / γ ray
→ Energy → MeV
→ Highly energetic photons
→ $\lambda \approx pm = 10^{-12}$ m

Measuring of radioactivity:

$A \longrightarrow B$

t N

t+dt N+dN

{N = No. of particles, dN = -ve}

* High Activity ⇒ Fastly A → B

No. of nuclei of A dissociated in this dt time interval = $-dN$

$$\text{Activity} = \frac{\text{No. of nuclei disintegrated}}{\text{Time taken}} = \frac{-dN}{dt}$$

$A = \left|\frac{dN}{dt}\right|$

Unit: (SI)

① dps = disintegration per sec = Becquerel = Bq

② 1 Curie = 1 ci = 3.7×10^{10} dps

③ 1 Rutherford = 1 Ru = 10^6 dps

→ All nuclei of A have equal probability to convert in B.
→ Which nucleus of A will convert into B is just a matter of chance.

Rutherford - Soddy law {Statistical law}:

$$A \longrightarrow B$$

	A (Unstable)	B (Stable)
$t=0$	N_0	0
$t=t$	N	N_0-N
$t=t+dt$	$N+dN$ $(dN=-ve)$	$N_0-(N+dN)$

* **Rate of disintegration $\propto$ No. of active Nuclei $\equiv \left|\frac{-dN}{dt}\right| \propto N$**

$\therefore$ * $\boxed{\frac{dN}{dt} = -\lambda N}$

λ = decay constant
- → Unit = sec^{-1}
- → Property of substance
- → doesn't depend on external parameters like Temp, pressure etc.

High Radioactive $\Rightarrow \lambda \uparrow$
Less Radioactive $\Rightarrow \lambda \downarrow$
No Radioactive | Stable $\Rightarrow \lambda = 0$

Now,

$$A \xrightarrow{\lambda} B$$

	A	B
$t=0$	N_0	0
$t=t$	N	N_0-N

$$\frac{dN}{dt} = -\lambda N$$

$$\int_{N_0}^{N} \frac{dN}{N} = -\lambda \int_0^t dt$$

$$(\ln N)_{N_0}^{N} = -\lambda t$$

$$\ln\left(\frac{N}{N_0}\right) = -\lambda t$$

$$\therefore \boxed{N_A = N = N_0 e^{-\lambda t}}$$

Now, $N_B = N_0 - N$

$$N_B = N_0 - N_0 e^{-\lambda t}$$

$$\boxed{N_B = N_0\{1 - e^{-\lambda t}\}}$$

Also, if N_A changes with time, so mass of A also changes with time.
Let at $t=0$, $m_A = m_0$

$$\boxed{m_A = m_0 e^{-\lambda t}}$$

Activity:

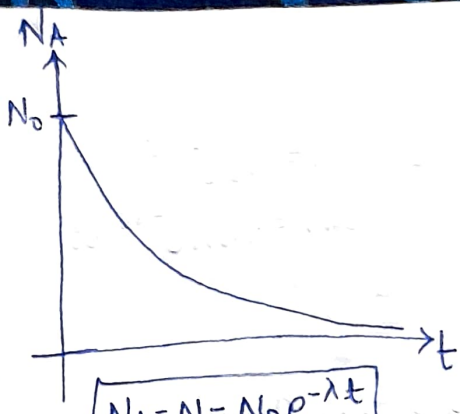
$$A = \left|\frac{-dN}{dt}\right| = \lambda N$$

$$\left|\frac{-dN}{dt}\right| = \lambda N_0 e^{-\lambda t}$$

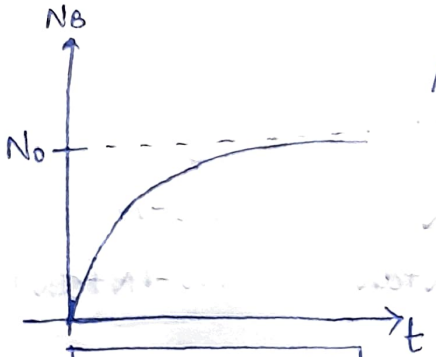
$$= \boxed{A = A_0 e^{-\lambda t}}$$

where $A_0 = \lambda N_0$

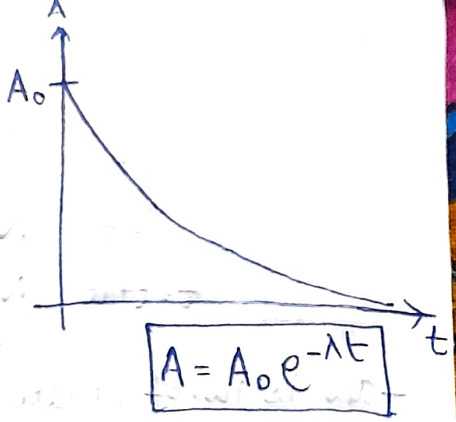
At $t=0$, $A = A_0 = \lambda N_0$



$$N_A = N = N_0 e^{-\lambda t}$$



$$N_B = N_0(1 - e^{-\lambda t})$$



$$A = A_0 e^{-\lambda t}$$

Half life ($t_{1/2}$): It is the time in which no. of active nuclei will become half of initial.

	A	B
t=0	N_0	0
t=t	N	$N_0 - N$

$$\text{At } t_{1/2} \Rightarrow N = \frac{N_0}{2}$$

At half life:

$$\frac{N_0}{2} = N_0 e^{-\lambda t_{1/2}}$$

$$= \ln\frac{1}{2} = -\lambda t_{1/2}$$

$$= \ln 2 = \lambda t_{1/2}$$

$$t_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$$

If Radioactivity $\uparrow \Rightarrow \lambda \uparrow \Rightarrow t_{1/2} \downarrow$

Note:

$$N_0 \xrightarrow{t_{1/2}} \frac{N_0}{2} \xrightarrow{t_{1/2}} \frac{N_0}{4} \xrightarrow{t_{1/2}} \frac{N_0}{8} \xrightarrow{t_{1/2}} \frac{N_0}{16} \cdots$$

So, let at $t=0$, $N_A = N_0$

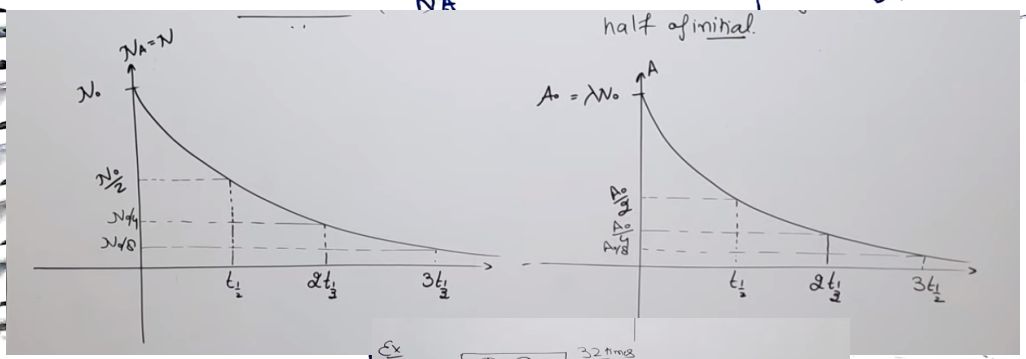
$$\Rightarrow \text{At } t = nt_{1/2},\ N_A = \frac{N_0}{2^n}$$

$$\therefore \star\ N_A = N = \frac{N_0}{2^{t/t_{1/2}}}$$

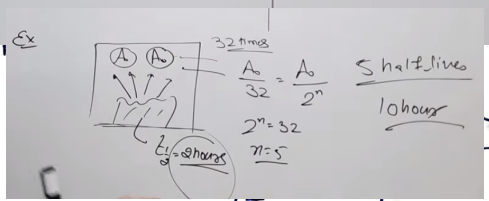
Similarly, at half life activity $A_0 \to A_0/2$

So, At $t = nt_{1/2}$; $A = \frac{A_0}{2^n}$

$$A = \frac{A_0}{2^{t/t_{1/2}}}$$



Ex



Mean life / Average life:

$$A \xrightarrow{\lambda} B$$

	A	B
$t=0$	N_0	0
$t=t$	N	N_0-N
$t=t+dt$	$N+dN$	$N_0-(N+dN)$

$N = N_0 e^{-\lambda t}$

$dN = -N_0 \lambda e^{-\lambda t} dt$

$-dN$ is no. of nuclei which lived for time t and then decayed.

$$t_{Avg} = \frac{\int (-dN)\, t}{N_0} = \frac{\lambda N_0 \int e^{-\lambda t} t\, dt}{N_0} = \lambda \int_0^\infty e^{-\lambda t} t\, dt$$

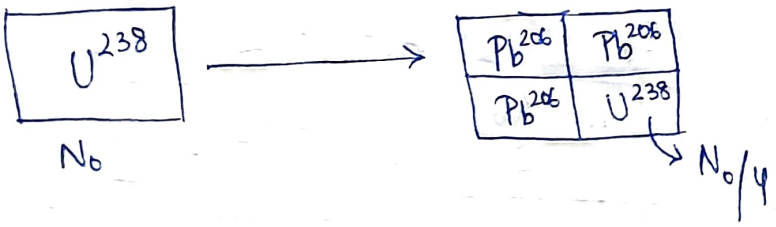
$$t_{Avg} = \lambda \left\{ -\frac{t}{\lambda} e^{-\lambda t} - \frac{1}{\lambda^2} e^{-\lambda t} \right\}_0^\infty = \left\{ -t e^{-\lambda t} - \frac{1}{\lambda} e^{-\lambda t} \right\}_0^\infty$$

$$t_{Avg} = \left\{ \frac{-\infty}{e^{\lambda \infty}} - \frac{1}{\lambda e^{\lambda \infty}} \right\} - \left\{ 0 - \frac{1}{\lambda} \right\} = \frac{1}{\lambda}$$

$$\therefore \boxed{t_{Avg} = \frac{1}{\lambda}}$$

Also, $\boxed{t_{1/2} = \frac{0.693}{\lambda} = 0.693\, t_{avg}}$

Q The number of U^{238} in an ancient rock is $\frac{1}{3}$rd of Pb^{206} atoms. The half life of U^{238} decay is 4.5×10^9 years. Assuming that Pb^{206} is formed by decay of U^{238}, find age of rock.



Since, $N_0 \to \frac{N_0}{4} \Rightarrow t = 2t_{1/2}$

$t = 2 \times 4.5 \times 10^9$

$t = 9 \times 10^9$ years

Q Cu^{64}: $\lambda = 1.516 \times 10^{-5}\ sec^{-1}$. Find activity of 1 mg of Cu^{64}.

$M_{Cu} = 63.5\ g/mol$

$$A = \lambda N = 1.516\times10^{-5} \times \frac{10^{-6}}{63.5} \times 6.023\times10^{23} = 1.43\times10^{11} \text{ dps}$$

$$A = \frac{1.43\times10^{11}}{10^6} = 1.43\times10^{5} \text{ Ru}$$

$$A = \frac{1.43\times10^{11}}{3.7\times10^{10}} = 3.86 \text{ Ci}$$

$$N = nN_A = \frac{m}{M}(N_A)$$

$$N = \frac{10^{-6}}{63.5} \times 6.023\times10^{23}$$

Q $t_{1/2}$ of Au^{198} is 2.7 days. Find ① decay constant ② Avg life ③ Activity of 1mg of Au $M_{Au} = 198$ g/mol

① $t_{1/2} = \frac{0.693}{\lambda} \Rightarrow \lambda = \frac{0.693}{t_{1/2}} = \frac{0.693}{2.7\times24\times60\times60} \text{ sec}^{-1} = 2.9\times10^{-6} \text{ sec}^{-1}$

② $t_{Avg} = \frac{1}{\lambda} = \frac{1}{2.9\times10^{-6}} \text{ sec}$ or $\frac{t_{1/2}}{t_{Avg}} = 0.693 \Rightarrow t_{Avg} = \frac{t_{1/2}}{0.693} = \frac{2.7}{0.693} = 3.9 \text{ days}$

③ $A = \lambda N = 2.9\times10^{-6} \times \frac{10^{-3}}{198} \times 6.023\times10^{23} = 8.8\times10^{17} \text{ dps}$

Q The activity of a radioactive sample falls from 600 dps to 500 dps in 40 minutes. Find its half life.

$A_0 = 600$ dps
$A = 500$ dps
$t = 40$ min

$A = A_0 e^{-\lambda t} \Rightarrow 500 = 600\, e^{-\lambda 40}$

$= \frac{5}{6} = e^{-\lambda 40} \Rightarrow \ln\frac{6}{5} = \lambda \times 40 \text{ min} \quad \text{—①}$

Also, $t_{1/2} = \frac{\ln 2}{\lambda} = \frac{\ln 2 \times 40 \text{ min}}{\ln 6/5} = 152 \text{ minutes}.$

Q Radon has $t_{1/2} = 4$ days, then find after how much time it will remain $\frac{1}{10}^{th}$ of initial.

Let at $t=0$, $N = N_0$

at $t=t$, $N = \frac{N_0}{2^{t/t_{1/2}}}$

now, $N = \frac{N_0}{10}$

$\Rightarrow \frac{N_0}{10} = \frac{N_0}{2^{t/t_{1/2}}}$

$\Rightarrow 2^{t/t_{1/2}} = 10$

$= \frac{t}{t_{1/2}} \ln 2 = \ln 10$

$t = t_{1/2} \frac{\ln 10}{\ln 2} = 4 \text{ days} \times \frac{2.303}{0.693}$

$t = 13.29$ days

Q In 5 sec, 1 gm of radioactive substance looses 1mg. Find $t_{1/2}$.

$m_0 = 1\,gm$

$m = 1\,gm - 1mg = 1gm - \frac{1}{1000}\,gm$

$m = 0.999\,gm$

$t = 5\,sec,$

Now, $m = \frac{m_0}{2^{(t/t_{1/2})}}$

$= 0.999\,gm = \frac{1\,gm}{2^{5/t_{1/2}}}$

$\Rightarrow 2^{5/t_{1/2}} = \frac{1000}{999}$

$= \frac{5}{t_{1/2}} \ln 2 = \ln\left(\frac{1000}{999}\right)$

$= \frac{5}{t_{1/2}} = \frac{\ln 1000 - \ln 999}{\ln 2}$

$= t_{1/2} = \frac{5 \ln 2}{\ln 1000 - \ln 999} = \frac{5 \times 0.693}{6.907 - 6.906} = 3465\,s$

Q A radioactive substance has initial activity 5000 dps. After 5 minutes, activity becomes 4000 dps. Find decay constant.

$A = A_0 e^{-\lambda t}$

$= 4000 = 5000\, e^{-\lambda(5\times 60)}$

$= \frac{4}{5} = e^{-300\lambda}$

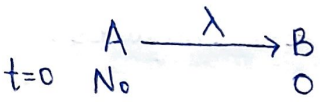
$= \frac{5}{4} = e^{300\lambda}$

$\Rightarrow 300\lambda = \ln\left(\frac{5}{4}\right)$

$\lambda = \frac{\ln(5/4)}{300}\,sec^{-1}$

$\boxed{\lambda = 0.0007438\,sec^{-1}}$

Probability:



t	No. of active nuclei of A	No. of nuclei of A dissociated	Probability of a nucleus of A to die $p = \frac{N_{decay}}{N_0}$	Probability of a nucleus of A to stay alive. $p = \frac{N_{alive}}{N_0}$
0	N_0	0	0	1
$t_{1/2}$	$N_0/2$	$N_0/2$	1/2	1/2
$2t_{1/2}$	$N_0/4$	$3N_0/4$	3/4	1/4
$3t_{1/2}$	$N_0/8$	$7N_0/8$	7/8	1/8
$nt_{1/2}$	$N_0/2^n$	$N_0 - \frac{N_0}{2^n}$	$\frac{N_0 - \frac{N_0}{2^n}}{N_0}$	$1 - \frac{\left(N_0 - \frac{N_0}{2^n}\right)}{N_0} = \frac{1}{2^n}$

At any time t,

Probability of a nucleus to stay alive $= \frac{N_{active}}{N_0} = \frac{N_0 e^{-\lambda t}}{N_0} = \frac{N_0}{2^{(t/t_{1/2})} N_0}$

$\therefore$ $\boxed{\text{Probability}_{alive} = e^{-\lambda t} = \frac{1}{2^{(t/t_{1/2})}}}$

Probability of a nucleus to decay $= 1 - e^{-\lambda t} = 1 - \frac{1}{2^{(t/t_{1/2})}}$

Q Factory $\xrightarrow{R}$ A (Radio Active) $\xrightarrow{\lambda}$ B (Stable)

Factory is manufacturing radioactive substance A at a rate 'R'. Decay constant of A is λ. Initially there are zero nuclei of A and B. At any time t :- ① Find no. of nuclei of A ② Find no. of nuclei of B.

① $\frac{dN_A}{dt} = +R - \lambda N_A$

$= \int_0^{N_A} \frac{dN_A}{R - \lambda N_A} = \int_0^t dt$

$= -\frac{1}{\lambda} \ln(R - \lambda N_A)_0^{N_A} = t$

$= \ln\left(\frac{R - \lambda N_A}{R}\right) = -\lambda t$

$\boxed{N_A = \frac{R}{\lambda}\{1 - e^{-\lambda t}\}}$

② $\frac{dN_B}{dt} = \lambda N_A$

$= \frac{dN_B}{dt} = \lambda \left\{\frac{R}{\lambda}(1 - e^{-\lambda t})\right\}$

$= \int_0^{N_B} dN_B = R\int_0^t (1 - e^{-\lambda t})dt$

$= N_B = R\left\{t + \frac{e^{-\lambda t}}{\lambda}\right\}_0^t$

$\boxed{N_B = Rt + \frac{R}{\lambda}\{e^{-\lambda t} - 1\}}$

Q Radioactive Series

	R.A. A	$\xrightarrow{\lambda_1}$	R.A. B	$\xrightarrow{\lambda_2}$	Stable C
t=0	N_0		0		0
t=t	N_A		N_B		N_C

At any time t, find no. of nuclei of ① A ② B ③ C

① $N_A = N_0 e^{-\lambda t}$

or

$\frac{dN_A}{dt} = -\lambda_1 N_A \Rightarrow \int_{N_0}^{N_A} \frac{dN_A}{N_A} = -\lambda_1 \int_0^t dt \Rightarrow N_A = N_0 e^{-\lambda_1 t}$

(2) $$\frac{dN_B}{dt} = \lambda_1 N_A - \lambda_2 N_B$$

$$\frac{dN_B}{dt} + \lambda_2 N_B = \lambda_1 N_0 e^{-\lambda_1 t}$$

I.F. $= e^{\lambda_2 t}$

$$e^{\lambda_2 t}\frac{dN_B}{dt} + \lambda_2 N_B e^{\lambda_2 t} = \lambda_1 N_0 e^{(\lambda_2-\lambda_1)t}$$

$$\frac{d}{dt}\{N_B e^{\lambda_2 t}\} = \lambda_1 N_0 e^{(\lambda_2-\lambda_1)t}$$

$$\int d(N_B e^{\lambda_2 t}) = \int \lambda_1 N_0 e^{(\lambda_2-\lambda_1)t} dt$$

$$N_B e^{\lambda_2 t} = \frac{\lambda_1 N_0 e^{(\lambda_2-\lambda_1)t}}{\lambda_2-\lambda_1} + C$$

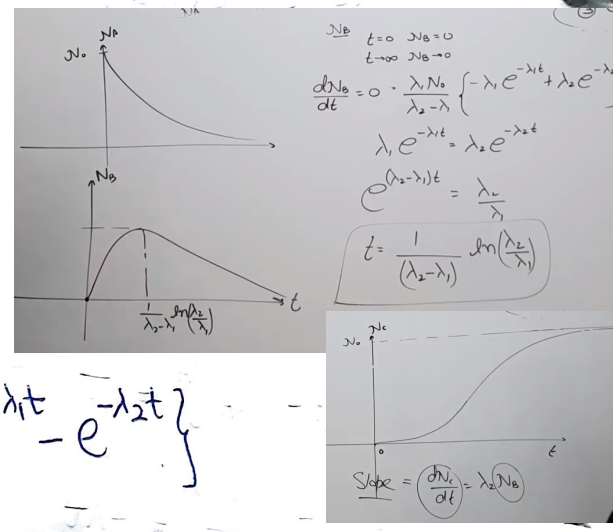
At $t=0$, $N_B = 0$

$$0 = \frac{\lambda_1 N_0}{\lambda_2-\lambda_1} + C$$

$$\boxed{C = -\frac{\lambda_1 N_0}{\lambda_2-\lambda_1}}$$

$$N_B e^{\lambda_2 t} = \frac{\lambda_1 N_0}{\lambda_2-\lambda_1} e^{(\lambda_2-\lambda_1)t} - \frac{\lambda_1 N_0}{(\lambda_2-\lambda_1)}$$

$$\boxed{N_B = \frac{\lambda_1 N_0}{\lambda_2-\lambda_1}\left\{e^{-\lambda_1 t} - e^{-\lambda_2 t}\right\}}$$



(3) $$\frac{dN_C}{dt} = \lambda_2 N_B$$

or

$$N_C = N_0 - N_A - N_B = N_0 - N_0 e^{-\lambda t} - \frac{\lambda_1 N_0}{\lambda_2-\lambda_1}\left\{e^{-\lambda_1 t} - e^{-\lambda_2 t}\right\}$$

Note: If anytime it is said that rate of formation and rate of decay is same then we say it is radioactive equilibrium

Q A solution contains a mixture of 2 isotopes A ($t_{1/2}$ = 10 days) and B ($t_{1/2}$ = 5 days). Initially total activity of solution is 10^{10} dps and after 20 days it becomes 2×10^9 dps. Find initial activity of A and B individually.

Activity: A, B

$t=0$ → x, $10^{10}-x$

$$A_A = \frac{A_0}{2^{t/t_{1/2}}} = \frac{x}{2^{20/10}} = \frac{x}{4}$$

$$A_B = \frac{10^{10}-x}{2^{20/5}} = \frac{10^{10}-x}{16}$$

We know, $$\frac{x}{4} + \frac{10^{10}-x}{16} = 2\times10^9 \text{ dps}$$

$$= 4x + 10^{10} - x = 32\times10^9$$

$$= 3x + 10^{10} = 3.2\times10^{10}$$

$$= 3x = 2.2\times10^{10}$$

$$\boxed{x = \frac{2.2}{3}\times10^{10}}$$

Q A piece of old wood has activity 1.95 dps per gram of ^{14}C. If fresh C^{14} (1 gm) has activity 15.6 dps, find age of the old wood. $T_{1/2}$ of C^{14} is 5570 years

$$C^{14} \longrightarrow C^{12}$$

$t=0 \quad A_0 = 15.6 \text{ dps}$

$$A = \frac{A_0}{2^{t/t_{1/2}}} \Rightarrow 1.95 = \frac{15.6}{2^{t/t_{1/2}}}$$

$$\Rightarrow 2^{t/t_{1/2}} = \frac{15.6}{1.95} = 8$$

$$\Rightarrow \frac{t}{t_{1/2}} = 3$$

$$t = 3 \times t_{1/2} = 3 \times 5570 \text{ yrs}$$

$$\boxed{t = 16710 \text{ years}}$$

Q A radioactive substance Na^{24} ($t_{1/2} = 15$ hrs) (activity = 1 μCi) is injected into the blood of a man. After 7.5 hours, a blood sample is taken from man, this sample has activity 4 dps. Find total volume of blood in man. Blood sample taken is $1 cm^3$

Let total volume of blood be V_0

$t=0, \quad A_0 = 10^{-6} Ci$

$= 10^{-6} \times 3.7 \times 10^{10} \text{ dps}$

$= 3.7 \times 10^4 \text{ dps}$

After 7.5 hrs, $A = 4$ dps for $1 cm^3$

$1 cm^3 \longrightarrow 4 \text{ dps}$

$V_0 \longrightarrow \frac{4 \text{ dps}}{1 cm^3} \times V_0 = A$

Now, $A = \frac{A_0}{2^{t/t_{1/2}}}$

$$\frac{4 \text{ dps}}{1 cm^3} V_0 = \frac{3.7 \times 10^4 \text{ dps}}{2^{7.5/15}}$$

$$\Rightarrow V_0 = \frac{3.7 \times 10^4 \times 1 cm^3}{4 \times \sqrt{2}}$$

$V_0 = 0.53 \times 10^4 cm^3 = 5.3 \times 10^3 cm^3$

$$\boxed{V_0 = 5.3 \text{ lt}}$$

Q A radioactive substance emit 1 β-particle on its decay. 2.3 mg of substance emit 10 β particles per second. $M_{subs} = 230 \text{ gm/mol}$. Find $t_{1/2}$ and decay constant of substance.

$A = 10 \text{ dps} = \lambda N$

$$10 = \lambda \left\{ \frac{2.3 \times 10^{-3} g}{230 g} \times 6.023 \times 10^{23} \right\}$$

$$10 = \lambda (6.023 \times 10^{18})$$

$$\lambda = \frac{10}{6.023 \times 10^{18}} = 1.66 \times 10^{-18} \text{ sec}^{-1}$$

$$t_{1/2} = \frac{0.693}{\lambda}$$

$$= \frac{0.693}{1.66} \times 10^{18}$$

$$= 4.175 \times 10^{17} \text{ sec}$$

Radioactive decay {displacement law}:

① α-decay:

→ It happens in heavier nuclei.

→ It increases ratio of $\frac{\text{no. of neutrons}}{\text{no. of protons}}$

Ex

$$_{92}U^{238} \longrightarrow {}_{90}Th^{234} + {}_{2}He^{4} + \text{Energy}$$

(α-particle)

$P = 92$

$n = 238 - 92 = 146$

$\frac{n}{P} = \frac{146}{92} = 1.586$

$P = 90$

$n = 234 - 90 = 144$

$\frac{n}{P} = \frac{144}{90} = 1.6$

Now, $\Delta E = (BE)_{product} - (BE)_{Reactant} = (B_{Th} + B_{He}) - (B_U)$

mass defect $= \Delta m = m_U - \{m_{Th} + m_{He}\} = \{M_U - 92m_e\} - \{M_{Th} - 90m_e + M_{He} - 2m_e\}$

$\Delta m = M_U - \{M_{Th} + M_{He}\}$

So, $\Delta E = \Delta mc^2 = \Delta m$ {in amu} $\times 931.5$ MeV

Now, α-decay;

Generally,

* → excited state

$$_{Z}X^{A} \longrightarrow {}_{Z-2}Y^{A-4*} + {}_{2}He^{4}$$

Parent nucleus — Daughter nucleus

$$_{Z-2}Y^{A-4*} \longrightarrow {}_{Z-2}Y^{A-4} + \gamma \text{ (gamma rays)}$$

{MeV}

$$\therefore \boxed{{}_{Z}X^{A} \longrightarrow {}_{Z-2}Y^{A-4} + {}_{2}He^{4} + \gamma + \Delta E}$$

$$\boxed{\Delta m = m_{Reactant} - m_{product} = M_X - \{M_Y + M_{He}\}}$$

$$\boxed{\Delta E = \Delta mc^2 = (M_X - M_Y - M_{He})\,931.5\,MeV = Q\text{-Value}}$$

Now,

$$X \longrightarrow Y + \alpha + \gamma + Q$$

γ-rays takes away energy $E = \frac{hc}{\lambda}$

Energy left $= Q - E$

(X) Rest $\xrightarrow{\text{breaks}}$ $K_Y \leftarrow$ (Y) $\quad P = \sqrt{2m_Y K_Y}$

$(\alpha) \rightarrow K_\alpha \quad P = \sqrt{2m_\alpha K_\alpha}$

E/C $\quad K_\alpha + K_Y = Q - E \quad —①$

M/C $\quad \sqrt{2m_Y K_Y} = \sqrt{2m_\alpha K_\alpha}$

$= \quad \frac{m_Y}{m_\alpha} = \frac{K_\alpha}{K_Y} \quad —②$

From ① and ②

$$K_Y = \frac{m_\alpha}{m_\alpha + m_Y}(Q-E) = \frac{4}{A}(Q-E)$$

$$K_\alpha = \frac{m_Y}{m_\alpha + m_Y}(Q-E) = \frac{A-4}{A}(Q-E)$$

these are rough calculations only since we have not considered the momentum of the emitted gamma ray

Q

$$_{92}U^{238} \longrightarrow {}_{90}Th^{234} + {}_{2}He^{4} + \gamma + Q$$

Atomic masses: 238.0508 amu $\quad 232.0377$ amu $\quad 4.0026$ amu

Find ① mass defect $= \Delta m = M_U - M_{Th} - M_{He} = 238.0508 - 232.0377 - 4.0026$

$\Delta m = 2.0105$ amu

② Q-value $= \Delta m \times 931.5\,MeV = 2.0105 \times 931.5\,MeV = 1872.78\,MeV$

③ If γ ray of $\lambda = 0.621$ pm is emitted then find KE of Th and KE of α-particle.

$$E_\gamma = \frac{hc}{\lambda} = \frac{12420\,eV\text{-}\mathring{A}}{0.621 \times 10^{-2}\mathring{A}} = \frac{12420 \times 100}{0.621}\,eV = \frac{12420}{0.621} \times 10^{-4}\,MeV = 2\,MeV$$

Energy left $= Q - E = 1872.78 - 2 = 1870.78$ MeV

$$K_\alpha = \frac{A-4}{A}(Q-E) = \frac{234}{238} \times 1870.78 = 1839.34\,MeV$$

$$K_{Th} = (Q-E) - K_\alpha = 1870.78 - 1839.34 = 31.44\,MeV$$

Note: If α-particle and daughter nucleus are having maximum KE, then γ-ray must not been emitted. $\therefore E_\gamma = 0$

$$K_\alpha = \frac{A-4}{A}Q$$

$$K_Y = \frac{4}{A}Q$$

$\rightarrow$ Max^m KE

Q $Th^{228} \longrightarrow Ra^{224} + He^{4} + Q$

Find ① Δm ② Q-value ③ K_{max} of Ra^{224} ④ K_{max} of α-particle

Atomic mass: 228.028726 amu, 224.020196 amu, 4.002600 amu

① $\Delta m = M_{Th} - M_{Ra} - M_{He} = 228.028726 - 224.020196 - 4.002600$

$\Delta m = 0.00593$ amu

② $Q = \Delta m c^2 = 0.00593 \times 931.5\ MeV = 5.52 MeV$

③ $K_{Ra} = \frac{4}{A} Q = \frac{4}{228} \times 5.52 = 0.0969\ MeV$

④ $K_{\alpha} = \frac{A-4}{A} Q = \frac{224}{228} \times 5.52 = 5.4231\ MeV$

② β-decay: Due to imbalance in nuclear and electrostatic forces, nucleus may get unstable and its one of neutron may convert into proton or one of proton may convert into neutron, while stabilising the nucleus. In this process they may emit β particles.

β-particles are of two types:

β^- particle	β^+ particle
→ electron	→ Positron
→ $_{-1}e^{0} = {}_{-1}\beta^{0}$	→ $_{+1}e^{0} = {}_{+1}\beta^{0}$
$Q = -1.6 \times 10^{-19}\ C$	$Q = +1.6 \times 10^{-19} C$
$m = 9.1 \times 10^{-31}\ kg$	$m = 9.1 \times 10^{-31}\ kg$
$E = mc^2 = 511\ KeV = 0.511 MeV$	$E = mc^2 = 0.511 MeV$

There are three types of β-decay:

(i) β^- decay: Electron is emitted

$$ {}_{Z}X^{A} \longrightarrow {}_{Z+1}Y^{A} + {}_{-1}e^{0} + Q $$

Parent — Daughter

Effectively, we can say that

$$ {}_{0}n^{1} \longrightarrow {}_{1}p^{1} + {}_{-1}e^{0} $$

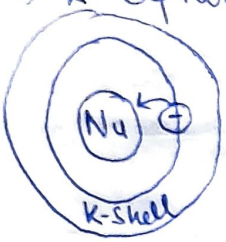
(ii) β^+ decay: Positron is emitted

1: $${}_{Z}X^{A} \longrightarrow {}_{Z-1}Y^{A} + {}_{+1}e^{0} + Q$$

Effectively, 2: $${}_{1}p^{1} \longrightarrow {}_{0}n^{1} + {}_{+1}e^{0} + Q$$

this effective reaction in isolation is not possible since mass of the RHS is more than the mass of the LHS
however, the rex. 1 is possible since mass defect is +ve

(iii) K-Capture:



$${}_{Z}X^{A} + {}_{-1}e^{0} \longrightarrow {}_{Z-1}Y^{A} + Q$$

K-shell $e^{\ominus}$

effectively ${}_{1}p^{1} + {}_{-1}e^{0} \rightarrow {}_{0}n^{1}$

Now,

(X) Parent —break→ (Y) K_Y ← (e) → K_e

mass defect = Δm
Q-value = Δmc^2

$$K_Y = \frac{m_e}{m_e+m_y} Q \approx 0 \quad (\text{as } m_y >>> m_e)$$

$$K_e = \frac{m_y}{m_e+m_y} Q \approx Q$$

E/C: $K_Y + K_e = Q$ —(1)

M/C: $\sqrt{2m_Y K_Y} = \sqrt{2m_e K_e}$

$$\frac{K_Y}{K_e} = \frac{m_e}{m_y} \quad -(2)$$

Experimental Findings:

(1) $K_e \leftarrow (e)$ (X) (Y) $\rightarrow K_Y$

$K_e + K_Y \neq Q$ ∴ E/C not valid

$K_e + K_Y < Q$

(2) After (X) breaks dirn of (Y) was not as expected.

M/C not valid

← (e) (X)

(X) → (Y)

(3) ${}_{0}n^{1} \longrightarrow {}_{1}p^{1} + {}_{-1}e^{0}$

$\pm\frac{1}{2}\frac{h}{2\pi}$ $\pm\frac{1}{2}\frac{h}{2\pi}$ $\pm\frac{1}{2}\frac{h}{2\pi}$

spin angular momentum

LHS ≠ RHS
as spin angular momentum not equal (total).

All these findings gave hint that there must some other particle emitted as well.

Pauli's Hypothesis:

→ Along with emission of $e^{\ominus}$, anti-neutrino ($\bar{\nu}$) will emit

→ Along with emission of e^{+}, neutrino (ν) will emit

neutrino and anti-neutrino have opposite spin-angular momentum (due to opposite directions of spin)

Neutrino / Anti-neutrino

- → Chargless
- → massless (very less mass) like photons
- → Energy packets
- → opposite spin angular momentum
- → Spin angular momentum = $\pm \frac{1}{2} \frac{h}{2\pi}$

I. β^- decay:

$$ {}_{Z}X^{A} \longrightarrow {}_{Z+1}Y^{A} + {}_{-1}\beta^{0} + \bar{\nu} + Q $$

Effectively, $$ {}_{0}n^{1} \longrightarrow {}_{1}p^{1} + {}_{-1}\beta^{0} + \bar{\nu} + Q $$

Mass defect = $\Delta m = m_x - \{m_y + m_e\}$

$= (M_x - Zm_e) - \{M_y - (Z+1)m_e + m_e\}$

$$ \boxed{\Delta m = M_x - M_y} $$ remember

$$ \boxed{Q = \Delta m c^2 = (M_x - M_y)\,931.5\ \text{MeV}} $$

Q is distributed among β^- and $\bar{\nu}$, since the Y nucleus is heavy hence its KE will be negligible

$$ \boxed{K_{\beta^- max} = Q} $$ when the anti-neutrino takes zero energy with it

Q

$$ {}_{Au}^{198} \longrightarrow {}_{Hg}^{198} + {}_{-1}e^{0} + \bar{\nu} + Q $$

Atomic mass	197.968233 amu	197.966760 amu

Find ① Mass defect = $M_{Au} - M_{Hg} = 197.968233 - 197.966760 = 0.001473$

② Q-value = $\Delta m c^2 = (0.001473)\,931.5\,\text{MeV} = 1.372\ \text{MeV}$

③ K_{max} of emitted electrons = $Q = 1.372\ \text{MeV}$

Ex $ {}^{19}O \longrightarrow {}^{19}F + {}_{-1}e^{0} + \bar{\nu} + Q $

19.003576 → 18.998403

find ① $\Delta m = m_o - \{m_F + m_e\} = (M_o - M_F) = 0.005173$

② Q value = 4.8186 MeV

③ max^m K.E. of emitted electrons = 4.8186 MeV

Ex $ {}^{23}_{10}Ne \longrightarrow {}^{23}_{11}Na + {}_{-1}e^{0} + \bar{\nu} + Q $

22.994466 → 22.989770

find ① $\Delta m = m_{Ne} - \{m_{Na} + m_e\} = \{M_{Ne} - M_{Na}\} = 0.004696$ amu

② Q value = $0.004696 \times 931.5 = 4.374$ MeV

③ max^m K.E. of emitted electrons = 4.374 Mev.

β^+ decay: $${}_{Z}X^{A} \longrightarrow {}_{Z-1}Y^{A} + {}_{+1}e^{0} + \gamma + Q$$

Effectively, $${}_{1}p^{1} \longrightarrow {}_{0}n^{1} + {}_{+1}e^{0} + \gamma + Q$$

Mass defect = $\Delta m = m_x - \{m_y + m_e\} = (M_x - Zm_e) - \{(M_y - (Z-1)m_e) + m_e\}$

$$\boxed{\Delta m = M_x - M_y - 2m_e}$$ remember

Q-value = $\Delta mc^2 = (M_x - M_y - 2m_e)931.5\,MeV = (M_x - M_y)931.5\,MeV - 2m_e(931.5)$

$$\boxed{Q = (M_x - M_y)931.5\,MeV - 2\times 0.511\,MeV}$$

better will be to remember that the mass defect corresponding to an e- will be = 511 keV = 0.511 MeV

sharing of the energy will be same as that in - β decay (Y will take negligible KE and all the energy will be shared b/w e+ and neutrino)

Q $$Al^{25} \longrightarrow Mg^{25} + {}_{+1}e^{0} + \gamma + Q$$

Atomic mass: Al^{25} 24.990432 amu; Mg^{25} 24.985839 amu

Find Q value of reaction.

Q value = $(M_{Al} - M_{Mg})931.5\,MeV - 2\times 0.511\,MeV$

$= (24.990432 - 24.985839)931.5\,MeV - 2\times 0.511\,MeV$

$= 4.2783 - 1.022\,MeV = 3.2563\,MeV$

K-capture:

$$ {}_{Z}X^{A} + {}_{-1}e^{0} \longrightarrow {}_{Z-1}Y^{A} + \gamma$$

Mass defect = $\Delta m = (m_x + m_e) - m_y = (M_x - Zm_e) + m_e - \{M_y - (Z-1)m_e\}$

$$\boxed{\Delta m = M_x - M_y}$$

$$\boxed{Q = \Delta mc^2 = (M_x - M_y)931.5\,MeV}$$

Q $$ {}_{26}Fe^{55} \longrightarrow {}_{25}Mn^{55}$$

Atomic mass: Fe^{55} 54.938298 amu; Mn^{55} 54.938050 amu

The above reaction can be carried by two ways ① β^+ decay ② K-capture

Find Δm and Q-value for both processes.

(1)

$$_{26}Fe^{55} \longrightarrow {}_{25}Mn^{55} + {}_{+1}e^{0} + \gamma$$

$\Delta m = (M_{Fe} - M_{Mn} - 2m_e) = (54.938298 - 54.938050 - 2\times\frac{1}{1800})$

$\Delta m = 0.000248 - 0.00111111$

$\Delta m = -0.00086$ amu {-ve indicates mass has increased}

$Q = \Delta mc^2 = -0.00086 \times 931.5$ MeV $= -0.8$ MeV {-ve indicates energy is to be provided}

(2)

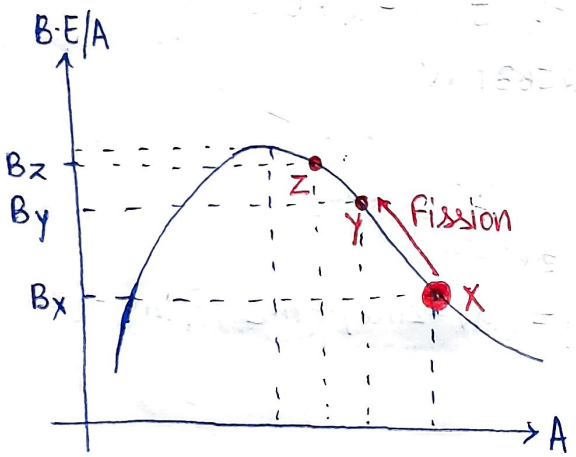
$$_{26}Fe^{55} + {}_{-1}e^{0} \longrightarrow {}_{25}Mn^{55} + \gamma$$

$\Delta m = (M_{Fe} - M_{Mn}) = 0.000248$ amu

$Q = \Delta m \times 931.5$ MeV $= 0.23$ MeV

∴ K-capture is more probable for this reaction.

Nuclear Fission:



$$X^{A_1} \longrightarrow Y^{A_2} + Z^{A_3} + \text{Energy}$$

Energy $= \Delta mc^2$

$= (m_x - m_y - m_z)c^2$

$= (M_x - M_y - M_z)c^2$

Energy $= (M_x - M_y - M_z)\,931.5$ MeV

Energy $= (BE)_{Product} - (BE)_{Reactant}$

$E = (B_y A_2 + B_z A_3) - B_x A_1$

→ Unstable heavier nuclei may dissociate/disintegrate into lighter nuclei and get more stable. Energy is also released in the process. This process is known as nuclear fission.

→ Spontaneous disintegration ⇒ Radioactivity

→ On emission of neutron/proton on heavy nucleus ⇒ Induced nuclear fission.

→ Mass will decrease

→ Energy released

→ Daughter nuclei are of comparable sizes,

→ These daughter nuclei may be in excited state and while coming in ground state, they may emit α, β and γ radiations.

Reactions in nuclear reactor:

induced nuclear reactions

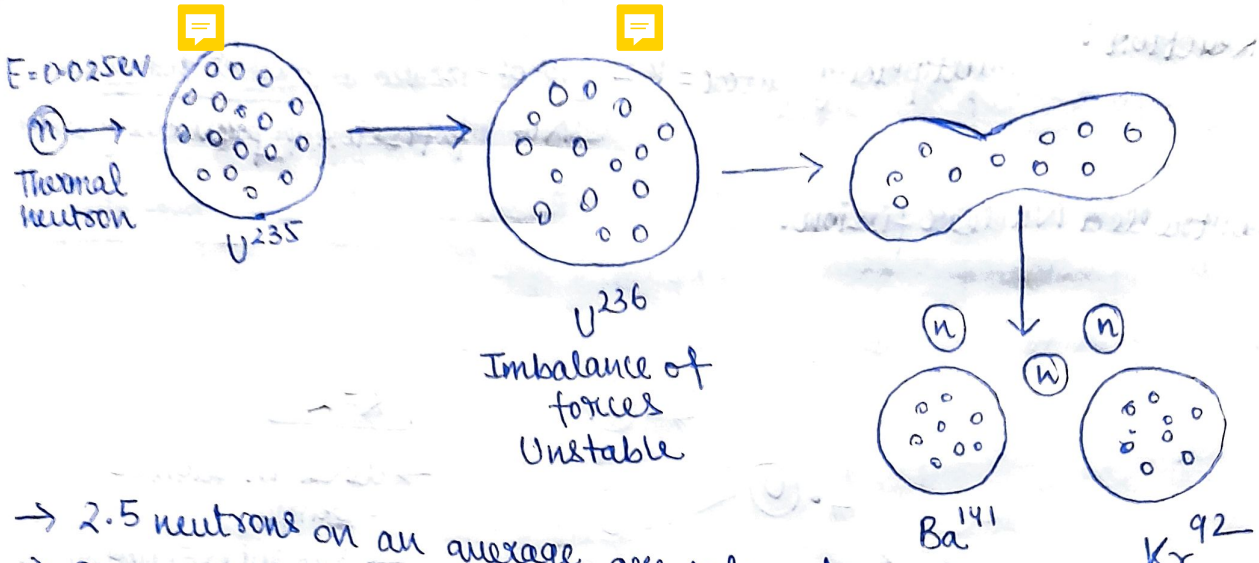
$$_{92}U^{235} + {}_0n^1 \longrightarrow {}_{92}U^{236} \longrightarrow {}_{56}Ba^{141} + {}_{36}Kr^{92} + 3\,{}_0n^1 \quad -(1)$$

(Thermal Neutron)

$E = 0.025\,eV$

$$\longrightarrow {}_{54}Xe^{140} + {}_{38}Sr^{94} + 2\,{}_0n^1 \quad -(2)$$

many other reactions are also possible



→ 2.5 neutrons on an average are released in one reaction.

→ Q_{Avg} of all reactions = 190 MeV

Ex Given atomic masses,

U^{236} = 236.045568 amu

Ba^{141} = 140.914400 amu

Kr^{92} = 91.926173 amu

Xe^{140} = 139.921641 amu

Sr^{94} = 93.915355 amu

n^1 = 1.008665 amu

Find mass defect and energy released in both reactions.

(1) $\Delta m = M_U - M_{Ba} - M_{Kr} - 3m_n$

$\Delta m = 236.045568 - 140.914400 - 91.926173 - 3 \times 1.008665$

$\Delta m = 0.179$ amu

$Q = \Delta mc^2 = 0.179 \times 931.5\,MeV = 166.74\,MeV$

(2) $\Delta m = M_U - M_{Xe} - M_{Sr} - 2m_n$

$\Delta m = 236.045568 - 139.921641 - 93.915355 - 2 \times 1.008665$

$\Delta m = 0.191$ amu

$Q = \Delta mc^2 = (0.191)\,931.5\,MeV$

$Q = 178.15\,MeV$

considering all the possible reactions, the average Q value of all possible reactions is 190 MeV

Note:

$$U \longrightarrow Ba + Kr + 3_0n^1$$

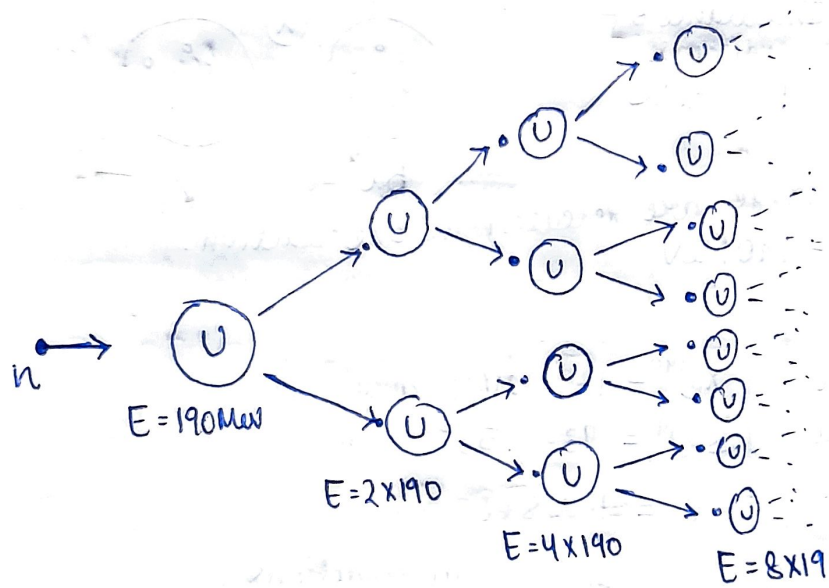
→ 80-85%. of energy is used by daughter nuclei in form of KE.
→ Neutrons use 3-5%. of energy
→ γ-radiations take up 2 to 4%. of energy
→ $Ba^* \longrightarrow Ba$ releases α, β radiations → 10%. energy

excited

Chain Reaction:

$$\text{Multiplicity factor} = K = \frac{\text{No. of fission in present generation}}{\text{No. of fission in previous generation}}$$

(i) Uncontrolled Nuclear fission:



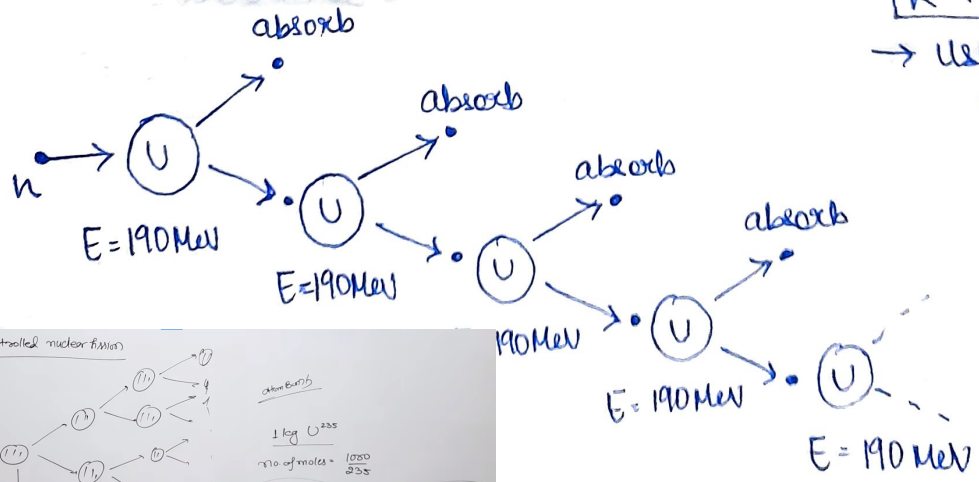
$K = 2$ = multiplicity factor (the ratio of U decayed in the current step to the previous step)

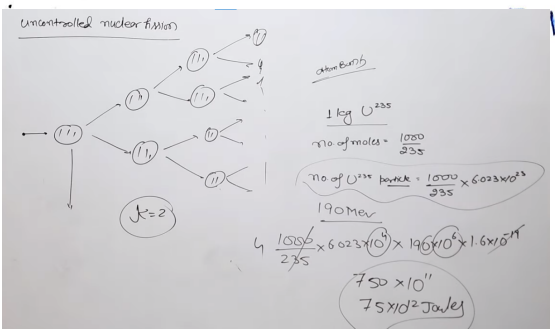
→ used in atomic bomb
→ produces enormous amount of energy rapidly.

(ii) Controlled Nuclear fission:

$K = 1$

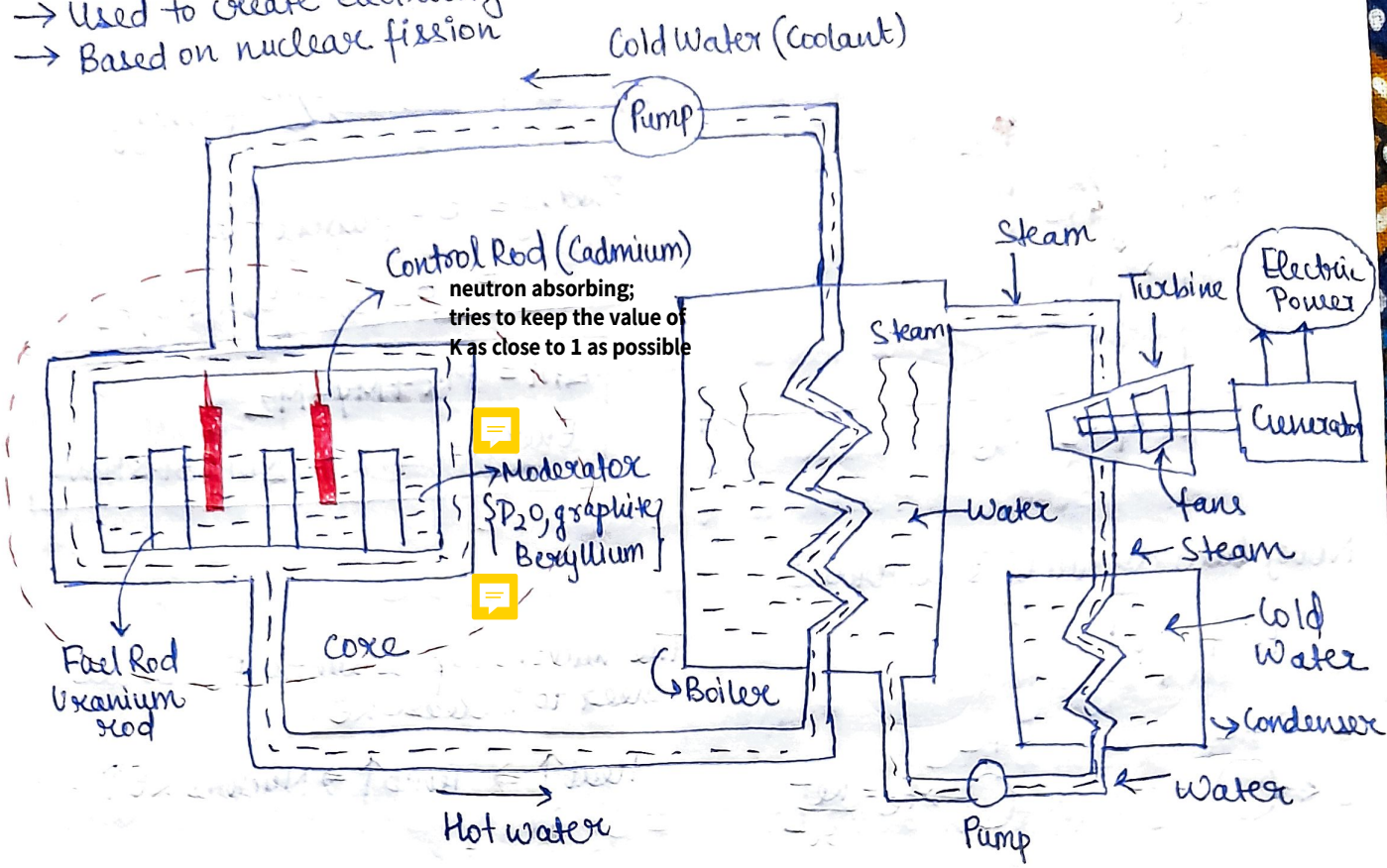
→ used in nuclear reactor





Nuclear Reactor:

→ Used to create electricity
→ Based on nuclear fission



→ Moderator is used to slow down neutrons by inelastic collision and decrease their KE from MeV to eV.

The neutrons having energies of the order of MeV do not collide with the Uranium atoms and instead pass through without causing any rex.

→ Control rod are used to absorb neutron

→ Efficiency $= \eta = \frac{E_{out}}{E_{in}} \times 100$

Q A nuclear reactor has Pout = 1000 MW and efficiency 20%. It uses U^{235} as fossil fuel. Fission of 1 U^{235} nucleus provides 200 MeV of energy. How many kgs of U^{235} is required by this reactor in year 2020.

$P_{out} = 1000 \times 10^6 = 10^9 W$

$\eta = \frac{P_{out}}{P_{in}} \times 100 \Rightarrow 20 = \frac{10^9}{P_{in}} \times 100$

$P_{in} = 5 \times 10^9 W$

$E = 5 \times 10^9 \times 366 \times 24 \times 60 \times 60 \ J$

Energy by 1 U^{235} fission $= 200 \times 10^6 \times 1.6 \times 10^{-19} J$

⇒ No. of U^{235} whose fission will take place in 2020:

$= \frac{5 \times 10^9 \times 366 \times 24 \times 60 \times 60}{200 \times 10^6 \times 1.6 \times 10^{-19}}$

No. of moles $=$

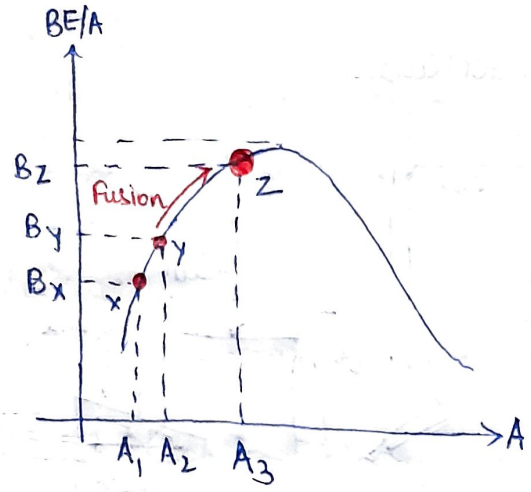
$= \frac{5 \times 10^9 \times 366 \times 24 \times 60 \times 60}{200 \times 10^6 \times 1.6 \times 10^{-19} \times 6.023 \times 10^{23}}$

Mass $= nM$

$= \frac{366 \times 6 \times 6 \times 6}{1.6 \times 6.023} \times \frac{235}{1000}$

$= 1927.83 \ kg$

Nuclear Fusion:



$$X^{A_1} + Y^{A_2} \longrightarrow Z^{A_3} + \text{Energy}$$

$$\text{Energy} = (BE)_{\text{product}} - (BE)_{\text{reactant}} = (B_Z A_3) - (B_X A_1 + B_Y A_2)$$

$$\Delta m = M_X + M_Y - M_Z$$

$$\text{Energy} = \Delta m c^2 = \Delta m \, 931.5 \text{ MeV}$$

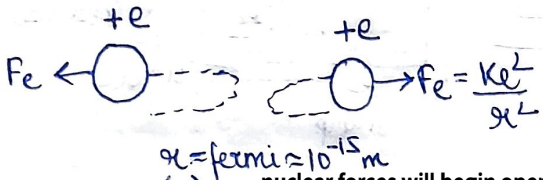
Now, how do nucleus combine?

$_1H^1 + {}_1H^1$

The nucleus repels due to Fe. So we need to increase KE.

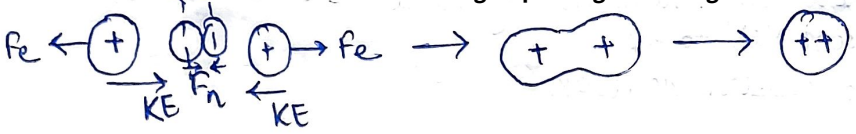
Heat ↑ ⇒ Temp ↑ ⇒ Nucleus KE ↑

as we provide more and more heat, the atom will get excited and the e- will leave the atom



r = fermi $\approx 10^{-15}$ m

nuclear forces will begin operating in this range



$$TMEC \Rightarrow 2KE + 0 = 0 + \frac{e^2}{4\pi\varepsilon_0 r} \Rightarrow KE = \frac{e^2}{4\pi\varepsilon_0 \times 2r} = \frac{(1.6\times10^{-19})^2 \times 9\times10^9}{2\times10^{-15}}$$

$$KE = \frac{2.56\times9\times10^{-14}}{2} \text{ J}$$

Also, $KE = \frac{3}{2}kT$ (k = Boltzman Const $= 1.38\times10^{-23}$ J/K)

$$T = \frac{2KE}{3k} = \frac{2\times2.56\times9\times10^{-14}}{2\times3\times1.38\times10^{-23}} \approx 10^9 \text{ Kelvin}.$$

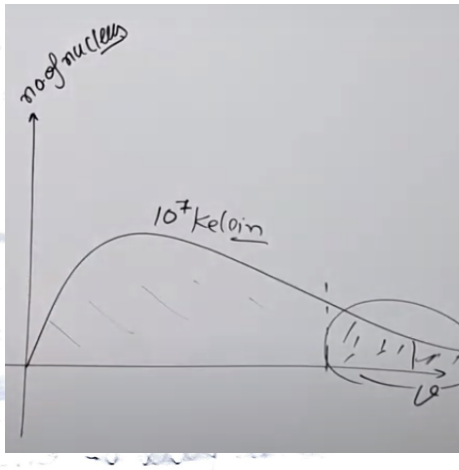
∴ For nuclear fusion temp should be in order of 10^9 Kelvin.

→ Nuclear fusion doesn't take place on earth naturally.

→ On earth, first we increase temp of the specific region using nuclear fission and when it reaches 10^9 Kelvin, then we carry out nuclear fusion.

Nuclear Fusion at Sun:

→ Sun is a large energy source
→ Energy is due to nuclear fusion
→ Temp of sun is 10^7 K but still nuclear fusion takes place as some particles have very high KE.



the predominant reactions taking place on the sun are:

① $_1H^2 + {}_1H^2 \longrightarrow {}_2He^3 + {}_0n^1 + Q$

② $_1H^2 + {}_1H^2 \longrightarrow {}_1H^3 + {}_1H^1 + Q$

③ $_1H^2 + {}_1H^3 \longrightarrow {}_2He^4 + {}_0n^1 + Q$

Ex Given, $_0n^1 = 1.008665$ amu $_1H^1 = 1.007825$ amu

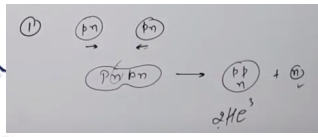
$_1H^2 = 2.014102$ amu $_1H^3 = 3.016049$ amu

$_2He^3 = 3.016029$ amu $_2He^4 = 4.002603$ amu

Find mass defect and Q value of each reaction

① $\Delta m = 2 \times 2.014102 - 3.016029 - 1.008665 = 0.00351$ amu

$Q = \Delta m \times 931.5$ MeV $= (0.00351)(931.5) = 3.26$ MeV

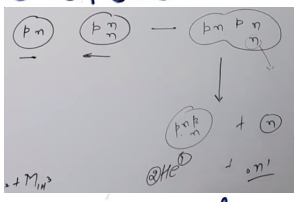


② $\Delta m = 2 \times 2.014102 - 3.016049 - 1.007825 = 0.00433$ amu

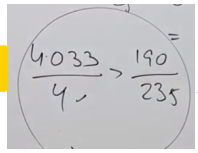
$Q = \Delta m \times 931.5$ MeV $= 0.00433 \times 931.5$ MeV $= 4.033$ MeV

③ $\Delta m = 2.014102 + 3.016049 - 4.002603 - 1.008665 = 0.018883$ amu

$Q = \Delta m \times 931.5$ MeV $= 0.018883 \times 931.5$ MeV $= 17.59$ MeV



Q $3\,{}_2He^4 \longrightarrow {}_6C^{12}$



Given, atomic mass of $_2He^4 = 4.002603$ amu

Find mass defect and Q-value of this reaction

$\Delta m = 3M_{He} - M_C$

$= 3 \times 4.002603 - 12.000000$

$= 12.007809 - 12.000000$

$= 0.007809$ amu

$Q = \Delta m \times 931.5$ MeV

$Q = 0.007809$ amu $\times 931.5$ MeV

$Q = 7.274$ MeV